

BUILDING A REASONING AND INFERENCING ENGINE FOR A STUDENT ACADEMIC ADVISOR

Abstract

This project presents the development of a rule-based expert system designed to function as an intelligent academic advisor for university students. The system uses Artificial Intelligence reasoning techniques to evaluate student academic records and generate recommendations based on predefined rules. Knowledge is represented using a JSON-based knowledge base containing academic policies. The system implements both Forward Chaining and Backward Chaining inference mechanisms to derive and verify conclusions. The final system demonstrates how expert systems can automate academic decision-making while providing transparent and explainable reasoning.

1. Introduction

One important branch of AI is Expert Systems, which use knowledge and inference rules to solve problems within a specific domain.

Academic advising is a process that requires evaluating various student attributes such as academic performance, attendance, disciplinary records, prerequisite completion, and financial status. Performing these evaluations manually can be time-consuming and prone to inconsistency.

This project develops a rule-based reasoning system capable of analyzing student information and generating academic recommendations automatically.

2. Problem Statement

Universities must continuously evaluate students to determine eligibility for scholarships, graduation, registration, and academic probation. Manual evaluation of these conditions can lead to delays and errors.

The challenge is to design an intelligent system that can:

- Represent academic knowledge in a structured format.
- Evaluate student records against institutional policies.
- Generate accurate academic recommendations.
- Explain how conclusions are reached.

3. Objectives

The objectives of this project are:

1. To represent academic advising knowledge using a structured knowledge base.
2. To develop a rule-based reasoning engine.
3. To implement Forward Chaining inference.
4. To implement Backward Chaining inference.
5. To generate explanations for system conclusions.

4. Knowledge Representation

The system uses a JSON knowledge base to store academic rules. Each rule consists of:

- Rule Name
- Conditions (IF part)
- Conclusion (THEN part)

Example Rule:

```
{  
  
  "name": "Scholarship Rule",  
  
  "if": {  
  
    "gpa": ">=3.5",
```

```
"attendance": ">80",  
"disciplinary": "No"  
},  
"then": "Eligible for Scholarship"  
}
```

The knowledge base contains rules for:

- Scholarship Eligibility
- Graduation Eligibility
- Academic Probation
- Registration Blocking
- Dean's List Eligibility

5. Reasoning and Inference Mechanisms

The core functionality of the system is based on rule-based reasoning. The reasoning engine uses a collection of facts and rules stored within a knowledge base to derive conclusions about a student's academic status.

The system employs two inference techniques:

1. Forward Chaining
2. Backward Chaining

These techniques enable the system to reason about student records and generate recommendations automatically.

5.1 Forward Chaining

Forward Chaining is a data-driven reasoning approach. The inference process begins with known facts about a student and progressively applies rules whose conditions match those facts.

In this project, the student's academic information acts as the initial set of facts. The reasoning engine examines each rule stored in the knowledge base and determines whether its conditions are satisfied.

When all conditions of a rule are satisfied, the rule is said to "fire," and its conclusion is added to the list of inferred outcomes.

The process continues until all rules have been evaluated.

This demonstrates how Forward Chaining starts with facts and discovers all possible conclusions.

```
FACTS:
GPA: 3.8
Attendance: 90
Disciplinary: No
Prerequisites: Completed
Fees: No
```

```
RULES TRIGGERED:
Scholarship Rule
Graduation Rule
Dean's List Rule
```

```
CONCLUSIONS:
Eligible for Scholarship
Eligible for Graduation
Dean's List Candidate
```

```
FACTS:
GPA: 2.4
Attendance: 60
Disciplinary: Yes
Prerequisites: Not Completed
Fees: Yes
```

```
RULES TRIGGERED:
Probation Rule
Registration Block Rule
```

```
CONCLUSIONS:
Academic Probation
Registration Blocked
```

5.2 Backward Chaining

Backward Chaining is a goal-driven reasoning approach. Instead of starting with facts, the process begins with a desired conclusion or goal.

The system attempts to determine whether sufficient evidence exists to support the selected goal.

To achieve this, the reasoning engine searches the knowledge base for a rule whose conclusion matches the desired goal. It then evaluates whether all conditions required by that rule are satisfied.

If all conditions are true, the goal is confirmed. Otherwise, the goal is rejected.

Example:

```
BACKWARD CHAINING DEMO  
  
GOAL: Eligible for Scholarship  
RULE: Scholarship Rule  
RESULT: True  
MESSAGE: Goal is satisfied
```

5.3 Comparison Between Forward and Backward Chaining

Although both techniques use the same knowledge base and rules, they differ in their reasoning strategies.

Forward Chaining begins with facts and searches for conclusions. It is useful when the objective is to discover all possible outcomes associated with a student's record.

Backward Chaining begins with a goal and searches for supporting evidence. It is useful when the objective is to verify a specific outcome such as scholarship eligibility or graduation eligibility.

In this project, both techniques produced consistent conclusions because they relied on the same set of rules and facts. However, they approached the reasoning process from different directions.

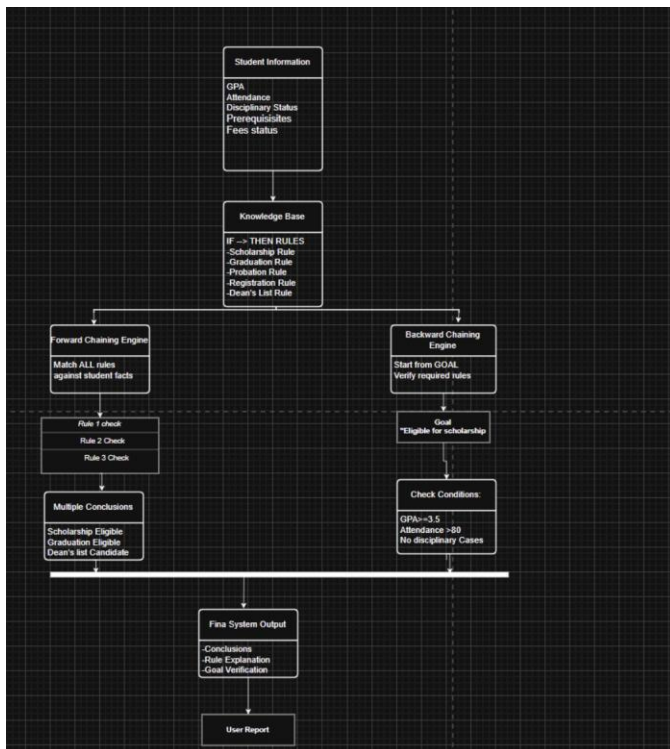
The implementation of both approaches demonstrates the flexibility of expert systems and highlights how different inference strategies can be applied to solve the same problem.

6. Inference Flow Diagram

The inference flow diagram illustrates how student facts are processed through the knowledge base and reasoning engines to generate conclusions.

The diagram includes:

- Student Facts
- Knowledge Base
- Forward Chaining Engine
- Backward Chaining Engine
- Rule Evaluation
- Final Conclusions



7 Conclusion

This project successfully implemented a rule-based expert system for academic advising. The system demonstrated key Artificial Intelligence concepts including knowledge representation, inference mechanisms, and explainable reasoning.

The use of JSON for knowledge representation provided flexibility, while the implementation of Forward Chaining and Backward Chaining allowed both discovery-based and goal-based reasoning. The project highlights the practical application of expert systems in educational environments.

References

Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* (4th Edition). Pearson.